Cloud Computing Course

Course n°5: Architecture on the Cloud

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Data Engineer
Roquette Frères 🌱

Quiz !

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Quiz

What is Git ?

A. A cloud platform for deploying applications
B. A continuous integration tool for testing code
C. A programming language for scripting
D. A version control system for tracking changes in code

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Quiz

What is GitHub Flow?

A. A type of continuous deployment strategy
B. A branching strategy for collaborative development on one branch
C. A branching strategy for collaborative development on multiples branches
D. A version control website for tracking changes in code

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Quiz

What is CI ?

A. The practice of automatically building and testing code when changes are made
B. A framework for monitoring live applications in production
C. A methodology for deploying code continuously
D. A tool for managing project tasks and issues

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Quiz

What is CD ?

A. A system for version-controlling code
B. A workflow for collaborating on Git branches
C. A testing methodology for verifying code quality
D. Automatically delivering or deploying code changes to production

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Reminder of previous course

CI/CD on the Cloud

- Reminder on Git
- Branching Strategies for development
- Continuous Integration
- Continuous Delivery / Deployment

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Contenu

Architecture on the Cloud

1. Title
2. Quiz
3. Certification
4. Security
5. Logging & Monitoring
6. Architecture

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Certification

Azure certification

AZ-900

Being certified is not mandatory to find a job, projects are more interesting. However, if you want to be certified, you can find 50% vouchers by taking an online Microsoft training course.
Example:

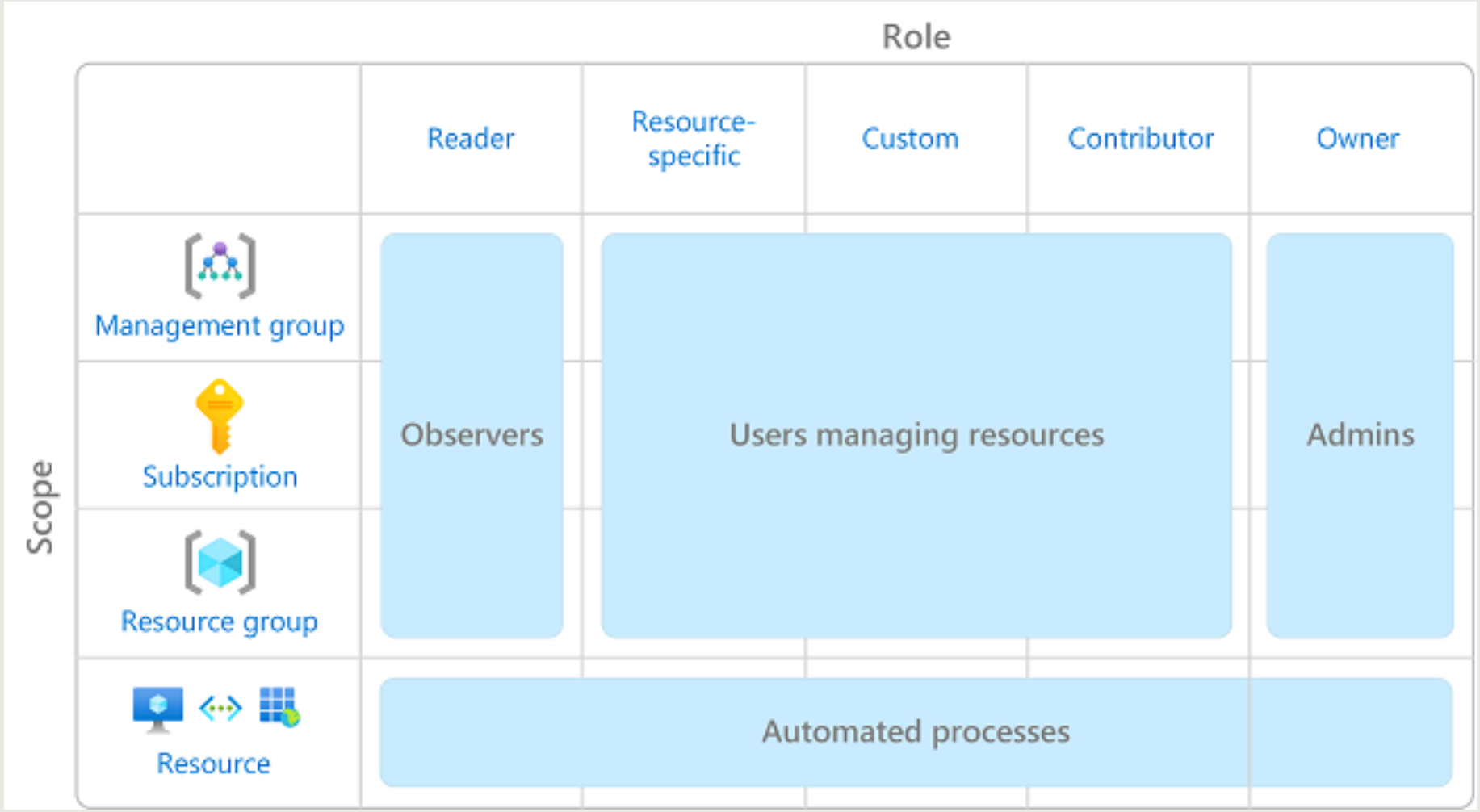
- Microsoft Azure Virtual Training Day: Fundamentals
- Microsoft Azure Virtual Training Day: Data Fundamentals

Apparently, by creating your Microsoft Learn account as a student, you can receive free voucher.

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Security

Reminder: Role-based access control (RBAC)



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Security

Secret Management

Secret Management

Types of secrets

- Credentials (username & password)
- API keys
- Connection string
- Client & secret keys

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Secret Management

Please I don't want to see that in any repo

Hard-coded secrets

```
db_password = "MyS3cr3tP@ssw0rd"
api_key = "sk_live_abc123xyz789"
```

The config .yaml file without env variables

```
database:
  password: "MyS3cr3tP@ssw0rd"
```

The .env in version control

```
DB_PASSWORD = "MyS3cr3tP@ssw0rd"
```


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Secret Management

Azure Key Vault

- Store secrets, key & certificate
- Encrypt at rest & in transit
- Versionize
- Monitor usage
- Can be accessed using managed identities

Azure Key Vault documentation



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Secret Management

```python
import os
from azure.keyvault.secrets import SecretClient
from azure.identity import DefaultAzureCredential

keyVaultName = os.environ["KEY_VAULT_NAME"]
KVUri = f"https://{keyVaultName}.vault.azure.net"

credential = DefaultAzureCredential()

# Create a client
client = SecretClient(vault_url=KVUri, credential=credential)

# Set a secert
secretName = 'MySuperSecret'
secretValue = '1234'

client.set_secret(secretName, secretValue)

# Retrieve a secret
retrieved_secret = client.get_secret(secretName)

# Delete a secret
poller = client.begin_delete_secret(secretName)
deleted_secret = poller.result()
```

Quickstart: Azure Key Vault secret client library for Python

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Secret Management

Note for the project

Since secrets cannot be stored directly in the code, please provide any required credentials or include them in the setup instructions within the documentation.

Ensure that you:

- Do not store secrets in the code
- Do not log secrets
- Separate secrets by environment

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Security

Policy

Azure Policy

What Is Azure Policy?

- A governance tool in Azure for creating, assigning, and managing policies
- Ensures resources stay compliant with organizational standards
- Uses policy definitions to enforce resource configurations

Azure Policy documentation

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Azure Policy

Note for the project

Policies can be defined using Terraform!

Policy Definition

- Rule describing allowed or denied behavior
- Example: *"Allowed locations"* or *"Require tags on resources"*

Policy Assignment

- Applying a definition to a scope (Subscription / Resource Group / Resource)

Policy Parameters

- Make policies reusable and customizable

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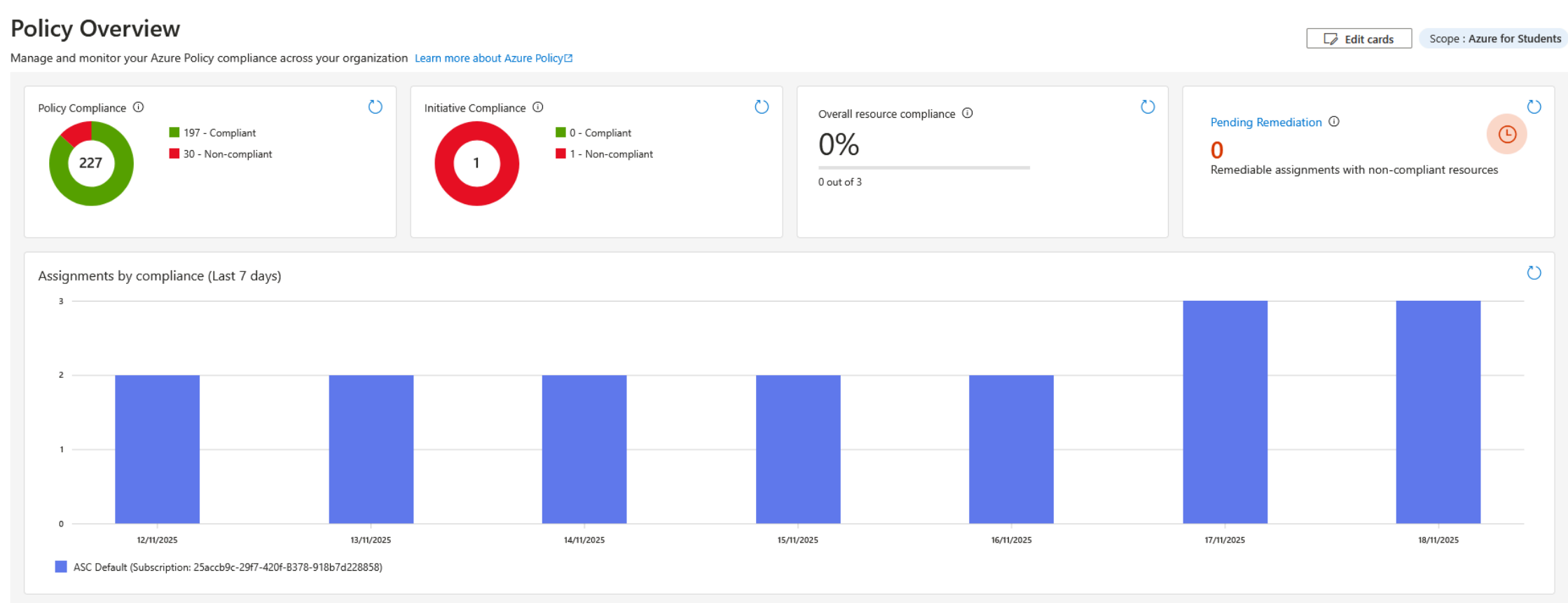
Azure Policy


```
{
  "properties": {
    "displayName": "Régions de déploiement de ressources autorisées",
    "policyType": "BuiltIn",
    "mode": "Indexed",
    "metadata": {
      "version": "1.0.0",
      "category": "System Policy"
    },
    "version": "1.0.0",
    "policyRule": {
      "if": {
        "allOf": [
          {
            "field": "location",
            "notIn": "['italynorth]"
          },
          {
            "field": "location",
            "notEquals": "global"
          },
          {
            "field": "type",
            "notEquals": "Microsoft.AzureActiveDirectory/b2cDirectories"
          }
        ]
      },
      "then": {
        "effect": "deny"
      }
    }
  },
  "id": "/providers/Microsoft.Authorization/policyDefinitions/b86dabb9-b578-4d7b-b842-3b45e95769a1/versions/1.0.0",
  "type": "Microsoft.Authorization/policyDefinitions/versions",
  "name": "1.0.0"
}
```


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Azure Policy

Dashboard



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Security

Compliance

Compliance

Definition

Compliance ensures that cloud resources follow organizational, legal, and regulatory requirements.

Examples

- GDPR (EU): General Data Protection Regulation
- HIPAA (US): Health Insurance Portability and Accountability Act
- PCI-DSS: Payment Card Industry Data Security Standard
- SOC 2: Service Organization Control 2
- ISO 27001: Information Security Management

Azure compliance documentation
Network and Information Security Directive (NIS2)

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Reminder: Azure Tags

Note for the project

You can define tags in your IaC scripts with Terraform!

Use tags on ressources to check

- Cost allocation -> Which team/project owns resources?
- Environment -> dev, staging, prod
- Compliance scope -> PCI, HIPAA, GDPR
- Data classification -> public, internal, confidential

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Monitoring and Logging

Monitoring and Logging

Why monitore & log applications

- Check system health
- Debug issues
- Optimize performance
- ...

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Azure Monitor

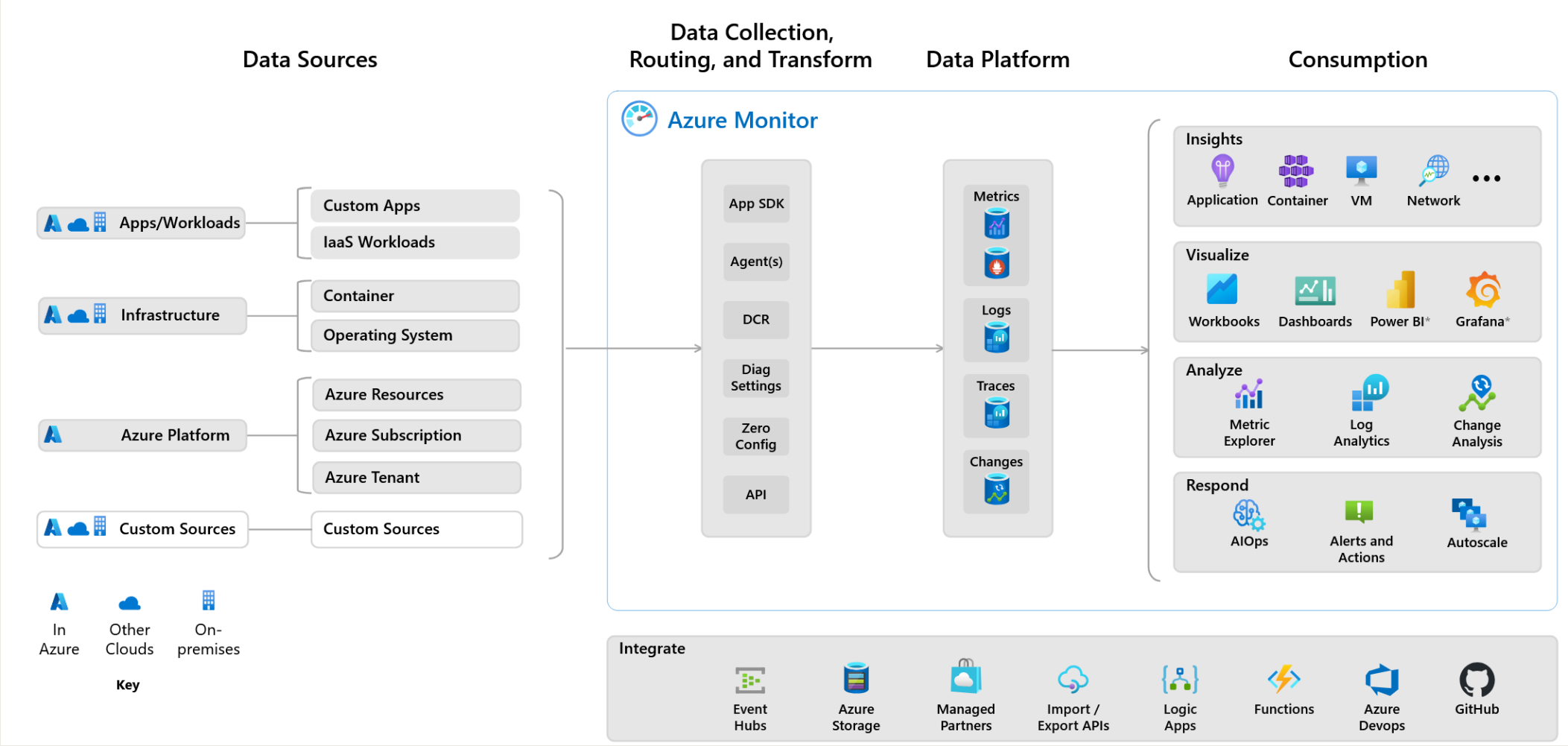
Azure Monitor is a unified platform to collect, analyze, and act on:

- Metrics
- Logs
- Traces
- Alerts
- Visualizations
- ...

Azure Monitor documentation

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Azure Monitor: Architecture



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Azure Monitor

Data Sources

Data sources are the types of resources being monitored.

- App: Application performance, health, and activity data
- Container: Data about containers and the applications running inside
- Operating system: Data about the OS on which your application is running
- Azure resource: Data about the operation of an Azure resource from inside the resource.
- Azure subscription: Operation and management of an Azure subscription
- And many more!

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Azure Monitor

Data Platform

- Metrics: Numeric measurements over time
- Logs: Discrete event records stored in Log Analytics
- Traces: Tracks operations across distributed systems
- Changes: Tracks configuration and resource changes

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Azure Monitor

Insights

Curated, resource-specific monitoring experiences with built-in dashboards:

- Application Insights
- Container Insights
- VM insights
- Network Insights

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Azure Monitor

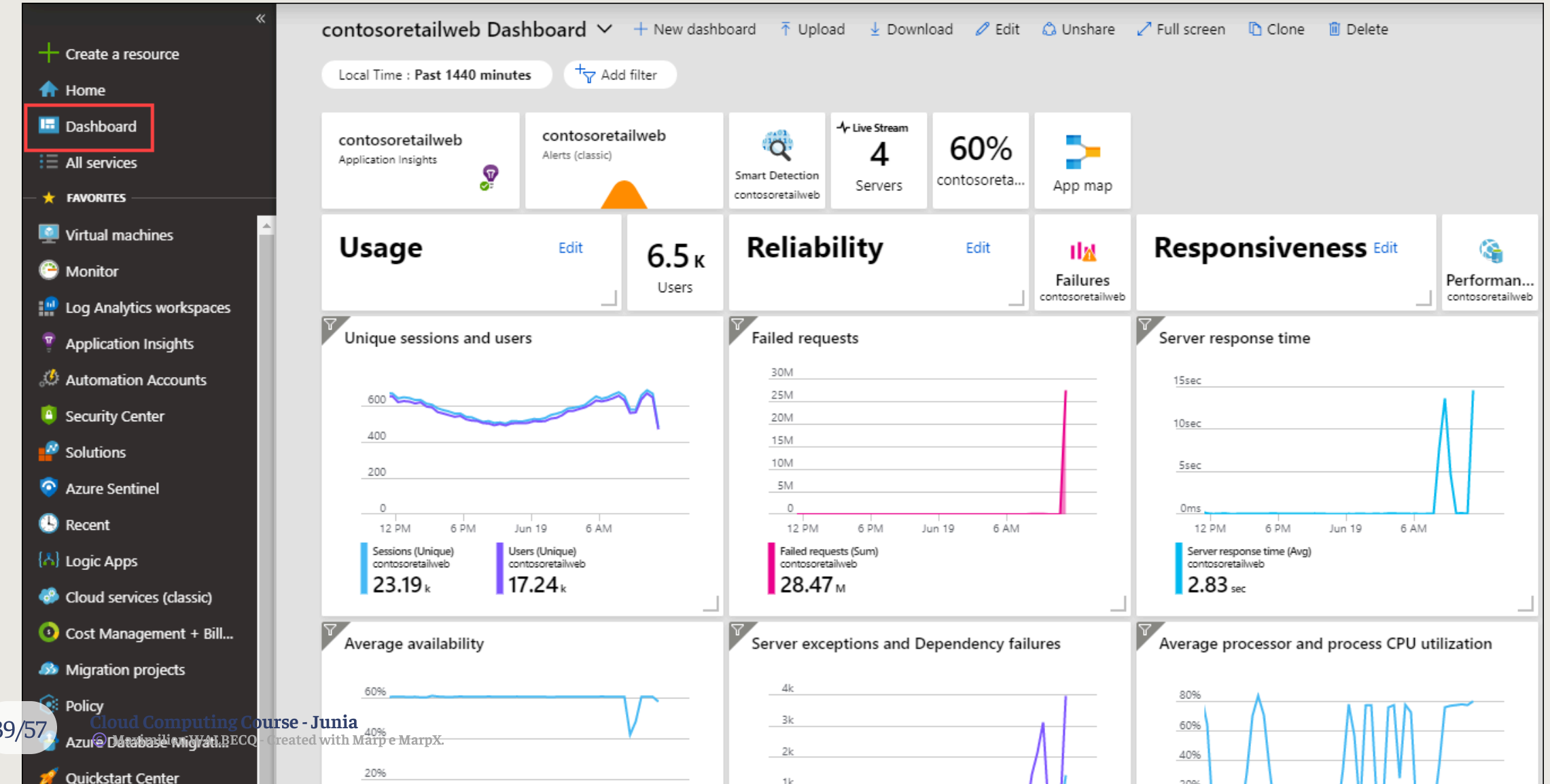
Visualize

Azure Monitor provides several visualization capabilities:

- Dashboard
- Workbooks
- Grafana Integration

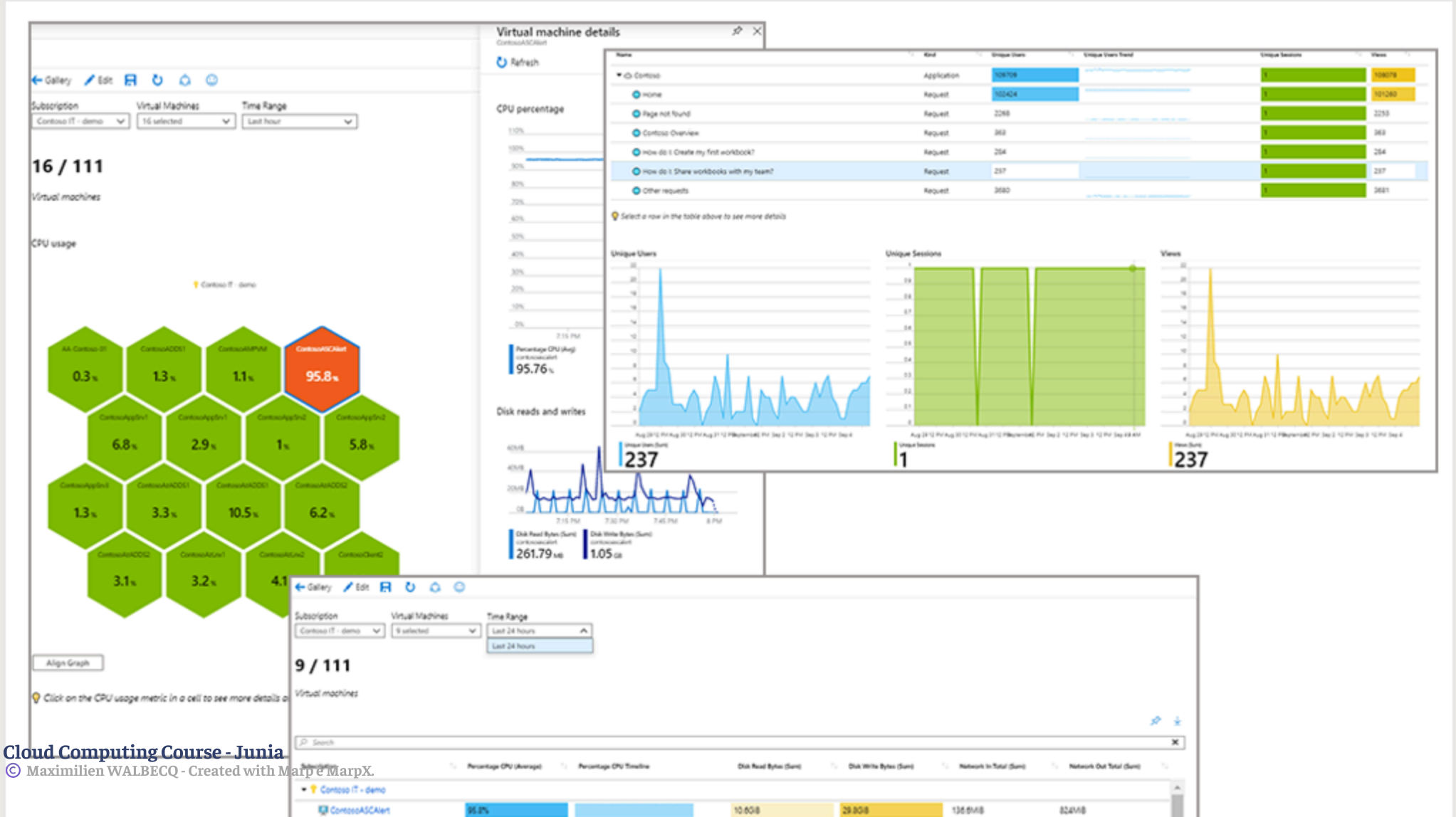
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Azure Monitor: Dashboard



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Azure Monitor: Workbook



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Azure Monitor

Analyze

- Metrics explorer
- Log Analytics
- Change Analysis

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Azure Monitor

Respond

Azure Monitor enables automated or manual responses when conditions occur.

Alerts

- Metric alerts: trigger on numeric thresholds
- Log alerts: trigger when query results meet conditions
- Activity log alerts: monitor management operations or policy changes

Action Groups

Execute actions such as:

- Email, SMS, push notifications
- Webhooks
- Azure Logic Apps
- Azure Functions

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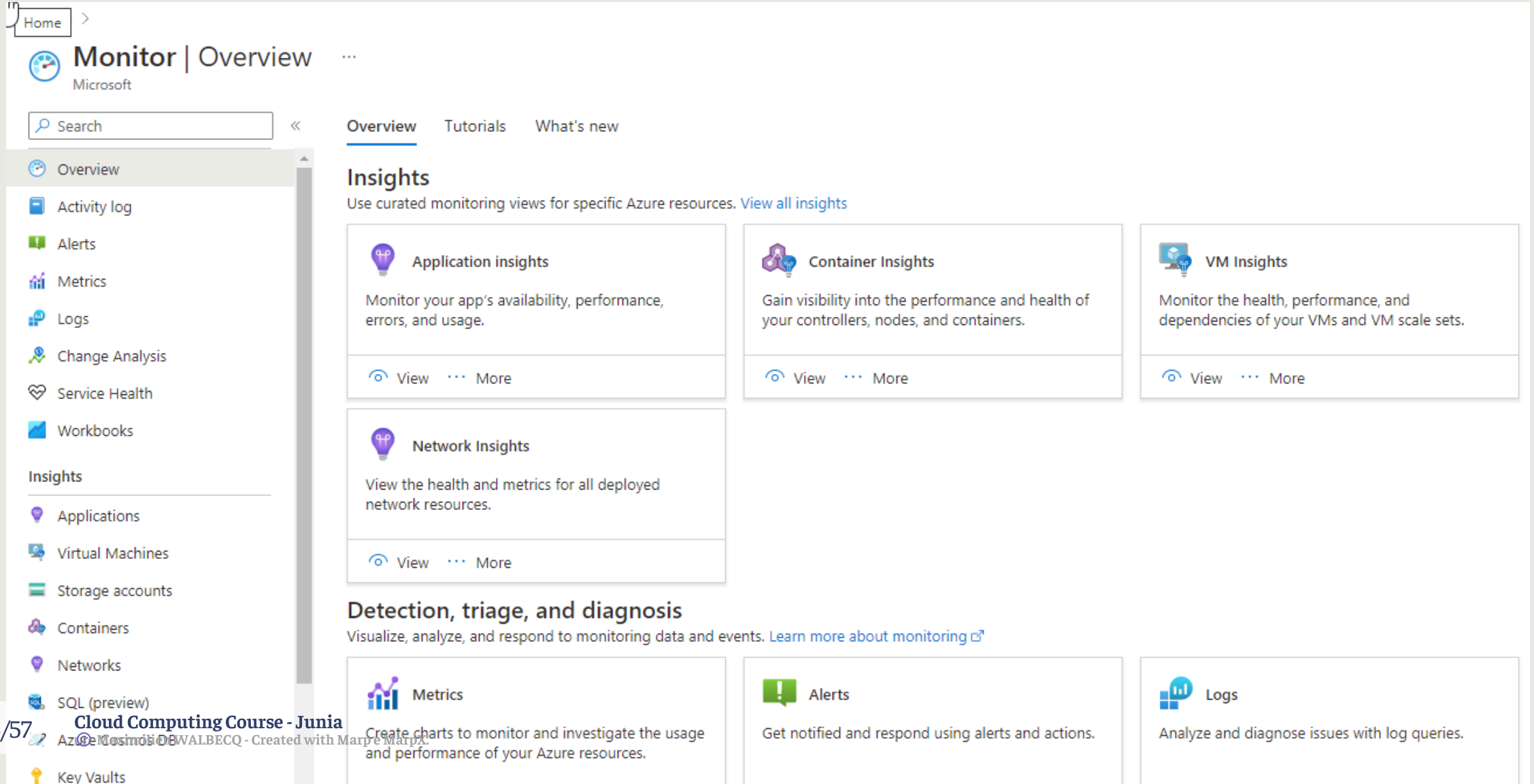
Azure Monitor

Integrate

- Event Hubs: Stream data
- Azure Storage: Long-term log archival
- Azure Logic Apps: Automated workflows
- Azure Functions: Serverless processing of alerts/events
- ...

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Azure Monitor: Portal



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Cloud Architecture

Reminder: The Twelve Factors App

It was created by "Heroku", a Platform as a Service cloud provider.
Source: https://12factor.net/

1. Codebase
2. Dependencies
3. Config
4. Backing Services
5. Build, Release, Run
6. Processes
7. Port Binding
8. Concurrency
9. Disposability
10. Dev/Prod Parity
11. Logs
12. Admin Processes

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Cloud architecture

Monolithic architecture

Definition

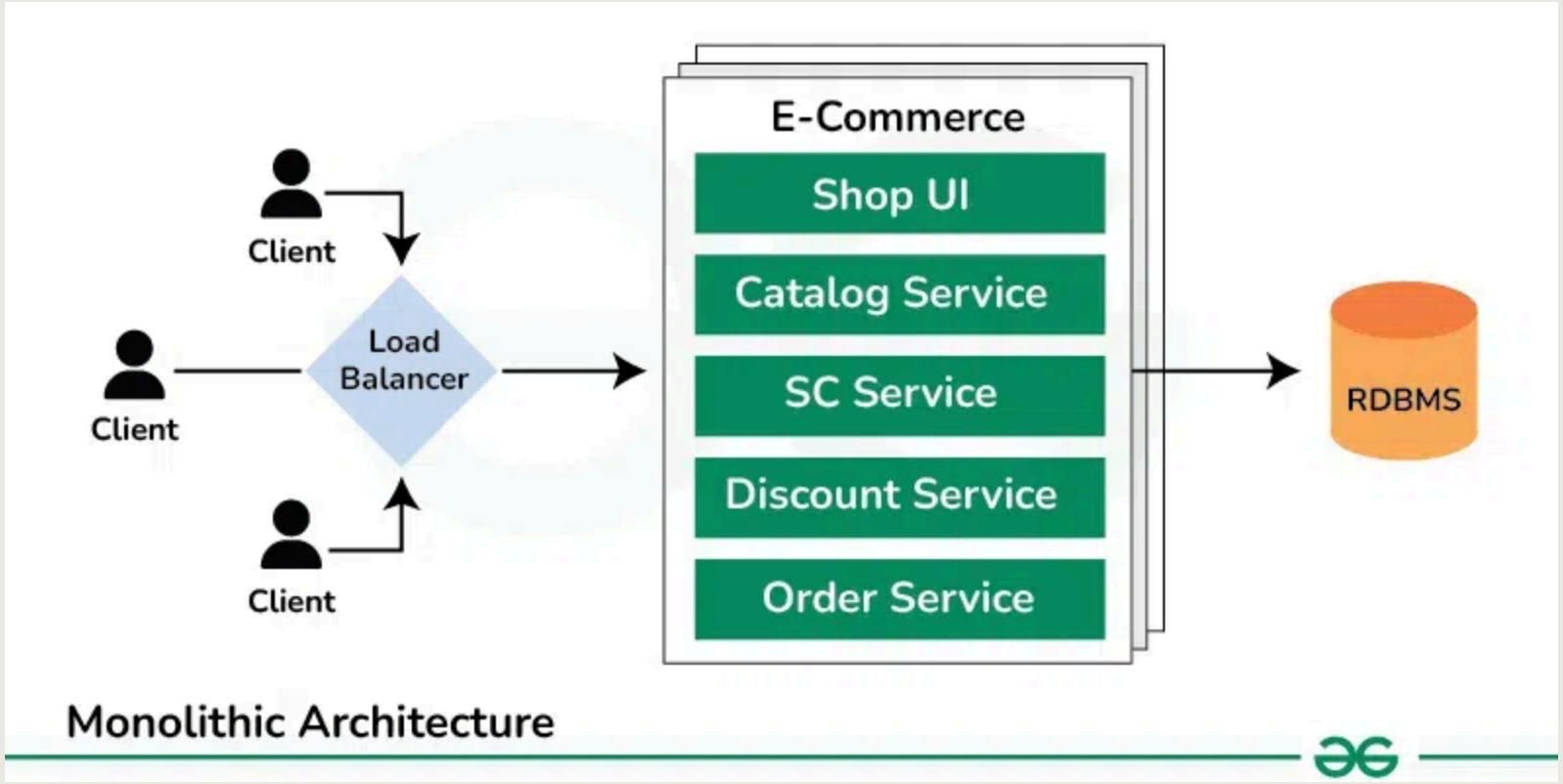
An architectural style where all components of an application are integrated into a single, unified codebase. The entire application is deployed as one executable or package.

Characteristics

- Single codebase: All modules are tightly coupled in one project
- Unified deployment: The entire application is built and deployed together
- Tight coupling: Changes in one part of the system may affect others
- Scaling limitations: Difficult to scale individual components independently

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Cloud architecture: Monolithic



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Cloud architecture

Microservices architecture

Definition

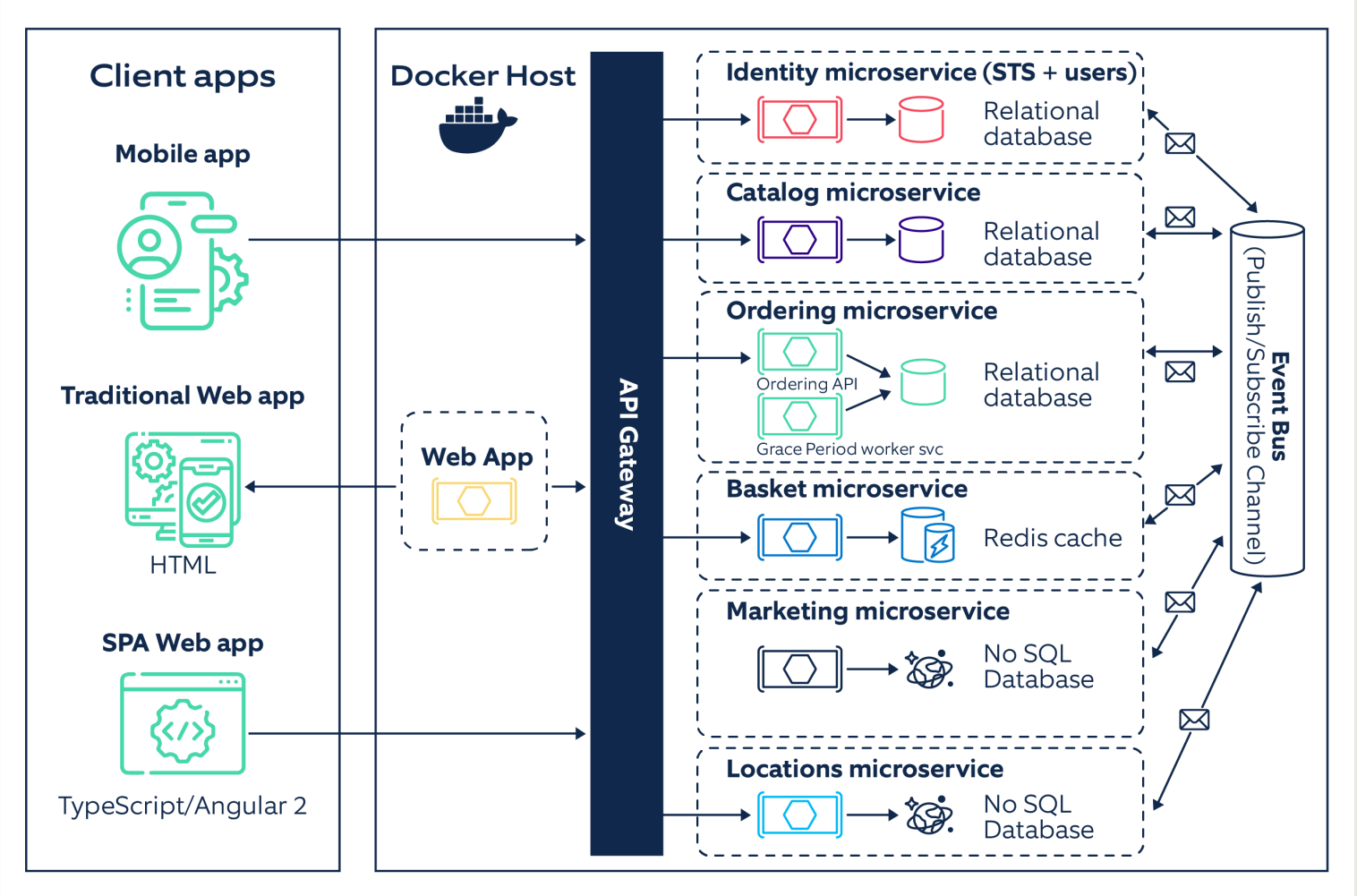
An architectural style where an application is composed of small, independent services that communicate

Characteristics

- Single Responsibility: Each service does one thing
- Independent Deployment: Deploy without affecting others
- Technology Agnostic: Use best tool for each service
- Decentralized Data: Each service owns its data
- Built for Failure: Resilience

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Cloud architecture: Microservices



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Cloud architecture

Monolithe & Microservices

- Monolithic architecture is generally simpler and easier to develop for small projects.
- Microservices architecture is better suited for applications with significant scaling or complexity needs.
- You don't have to choose 100% monolithic or 100% microservices, you can go hybrid.

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Cloud architecture: Microservices

Service Communication: Synchronous

Definition

- REST API: Services communicate by sending direct HTTP requests and waiting for a response.
- SOAP: Other protocol for synchronous communication.

Characteristics

- Real-time interaction between services.
- Easy to understand and implement.
- Latency or failure in one service can block others.

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Cloud architecture: Microservices

Service Communication: Asynchronous

Definition

- Event-driven architecture: Services communicate by sending events/messages
- Event Producer: Publishes events
- Event Consumer: Subscribes to and processes events
- Event Channel: Delivers events (ex: message queue)

Characteristics

- Services operate independently and respond when ready.
- Improved scalability and resilience.
- Can handle high throughput without blocking services.
- More complex to implement and monitor.

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Event-driven architecture style

Cloud architecture

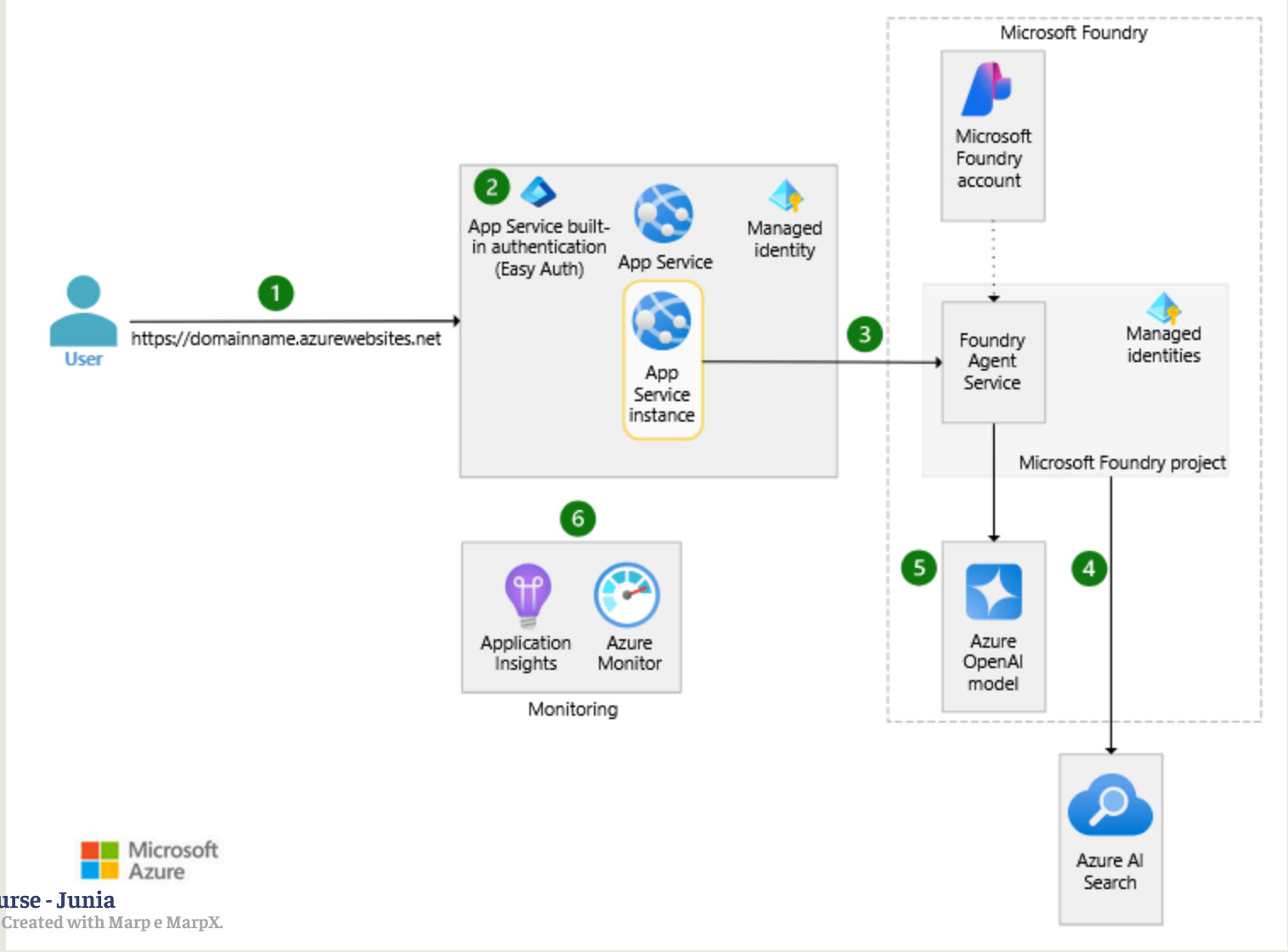
You can find many architecture examples online if you need help

Search on Azure Architectures for:

- Architecture schemas
- Steps by steps explanation
- Repositories with examples

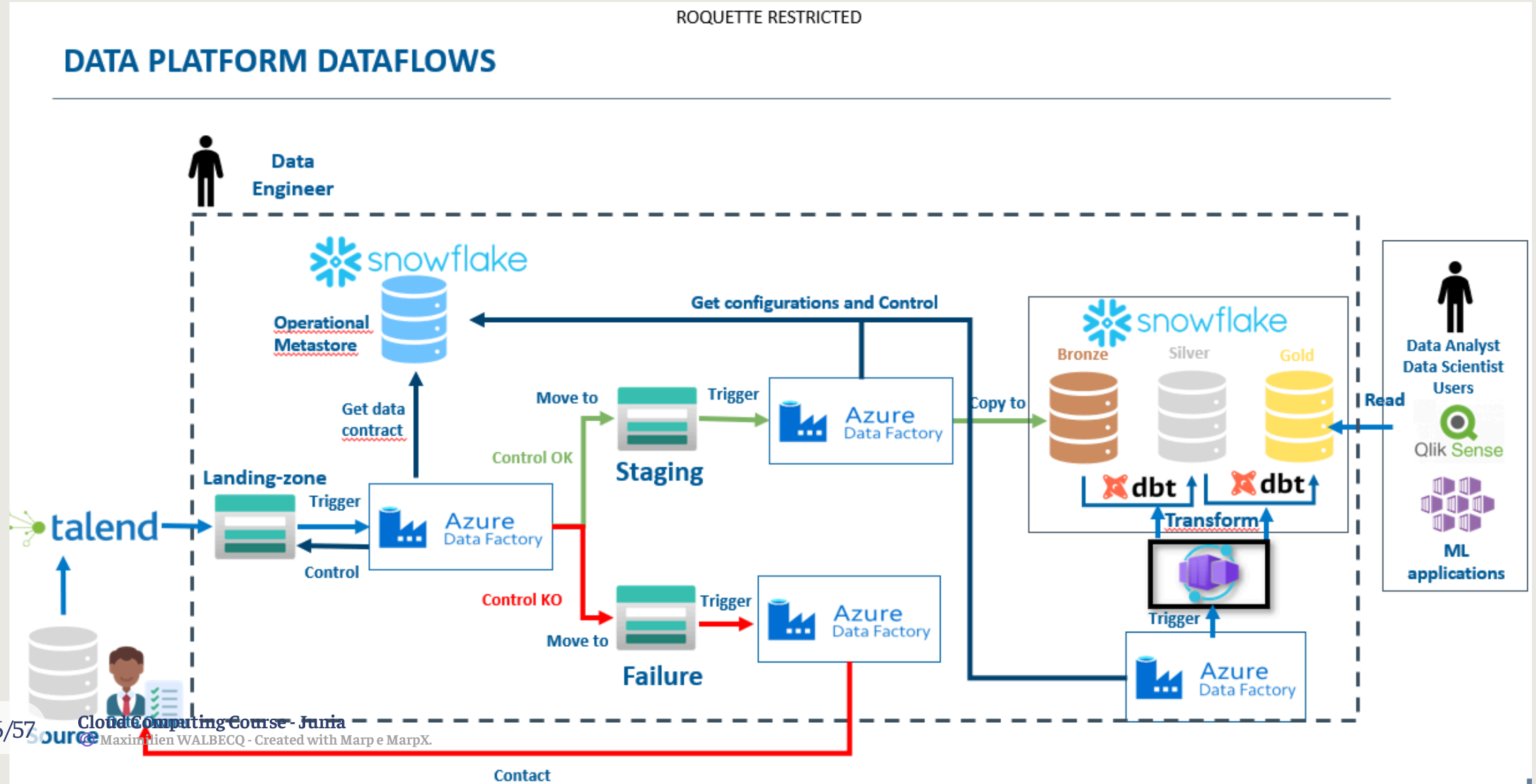
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Cloud architecture: Example 1



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Cloud architecture: Example



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Thank you !

Let's take a break!